

IPL related

version 1.0

1. IPL for different memories

Storage	Base name
SPI NOR/NAND	IPL and IPL_CUST files are required
EMMC	Just use IPLX_EMMC

2. IPL parameter description

When the chip uses an external DDR, the correct IPL should be selected according to the actual DDR parameters, otherwise the DDR may fail to initialize and not work properly. The following table describes the common parameters of IPL in the SDK package we release:

Parameter	Description
1600_4X 1866_4X 2133_4X	DDR speed grade, without these parameters the default is 2133
NO_JTAG	If the jtag version is not enabled, without this parameter, jtag is enabled by default
ext_NANYA_DDR3_6L	The version suitable for 6-layer board, without this parameter, it is suitable for 4-layer board by default

3. Special editions and supplementary instructions

Parameter	Description
IPL.LoadFromSD.bin IPL_CUST.LoadFromSD.bin	The version that supports booting from SD card, please use these two files when you need to start IPL and Uboot from SD card

Additional instructions:

- Although there is the word nanya in the IPL parameters, the mainstream DDR can basically be used
- The jtag pin may be multiplexed into other functions (I2C, etc.) during hardware design. In this case, the NO_JTAG version needs to be used, otherwise the function will not work properly

4. How to check the DDR Phase parameter setting in SSD26x IPL

After the machine is turned on, read 0x1010 Bank and compare the registers for confirmation. The following is the DDR Phase parameter/Bank 1010 register setting in the latest IPL of SSD26x (21-09-07): focus on checking the content in the red box; this setting is compatible with 2133 & 1866 rate (that is, the usage of 1866 is to slow down to 1866 on the setting of 2133)

4L Laminate Design:

SigmaStar Controller

Read Bank

VD Bank (MCU 32)

XDATA

Color

Bank (MCU32)

Bank: 0x1010

02~3D

1002~1016

1017~102B

102C~103D

☐ MENULOAD
☐ USB
☐ DSCRGB
☐ MIU

☐ VD_MHEG5
☐ ISP
☐ PM_SLP
☐ VD_MCU

☐ DDC
☐ BDMA
☐ MHEG5
☐ MVOP

☐ POR_STATU
☐ RVD
☒ MCU
☐ TSP0

☐ MIU2
☐ CLK_GEN
☐ MVD
☐ TSP1

Addr Unit

☐ 8 bit
☒ 16 bit

Clear

Address

00

Value

0x 0000

(Dec) 0

Bit Edit

15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

□ □ □ □ □ □ □ □ □ □ □ □ □ □ □

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

□ □ □ □ □ □ □ □ □ □ □ □ □ □ □

	00 / 08	01 / 09	02 / 0A	03 / 0B	04 / 0C	05 / 0D	06 / 0E	07 / 0F
00	0000	AAAA	0008	0000	003F	3300	0000	009B
08	0000	0000	0000	0200	0000	8333	0020	6005
10	4020	0004	0000	2222	8000	0000	0114	3232
18	67D7	001A	2004	0400	2277	7066	9322	B133
20	0000	0000	0000	0800	3277	7036	0022	0033
28	1122	0066	C000	0000	0A0A	AAAA	AAAA	AAAA
30	B3C8	007F	F000	0029	2020	2020	0808	0808
38	0800	8801	0707	0707	0001	0000	0000	0000
40	0000	0000	0000	0000	0011	0001	0000	0000
48	F0F1	0000	1414	1215	0000	0000	FF00	007F
50	0121	1111	2130	4221	2120	3333	4021	2131
58	3802	000E	0000	0000	3333	0333	0333	0333
60	001F	0000	0000	0000	0000	0000	0000	0000
68	3555	5555	4444	4444	0045	0000	0000	0000
70	5555	5555	4444	4444	0045	0000	0000	1F1F
78	1F1F	0000	0000	0200	8600	0007	C0C0	C0C0

6L Laminate Design:

SigmaStar Controller

Read Bank		VD Bank (MCU 32)		HK FRC		DMD		
XDATA		Color		Bank (MCU32)				
Bank: 0x1010								
02~3D		1002~1016		1017~102B		102C~103D		
☐ MENULOAD		☐ USB		☐ DSCRGB		☐ MIU		
☐ VD_MHEG5		☐ ISP		☐ PM_SLP		☐ VD_MCU		
☐ DDC		☐ BDMA		☐ MHEG5		☐ MVOP		
☐ POR_STATU		☐ RVD		☒ MCU		☐ TSP0		
☐ MIU2		☐ CLK_GEN		☐ MVD		☐ TSP1		
Addr Unit								
☐ 8 bit								
☒ 16 bit								
Clear								
Address		Value						
00		0x 0000						
		(Dec) 0						
Bit Edit								
15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0								
31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16								
	00 / 08	01 / 09	02 / 0A	03 / 0B	04 / 0C	05 / 0D	06 / 0E	07 / 0F
00	0000	AAAA	0008	0000	003F	3300	0000	009B
08	0000	0000	0000	0200	0000	8333	0020	6005
10	4020	0004	0000	2222	8000	0000	0114	3333
18	A229	001D	2004	0400	1077	103D	9322	B133
20	0000	0000	0000	0800	1077	101D	0022	0033
28	1122	008A	C000	0000	0A0A	AAAA	AAAA	AAAA
30	B3C8	007F	F000	002D	2020	2020	0808	0808
38	0800	8801	0707	0707	0001	0000	0000	0000
40	0000	0000	0000	0000	0011	0801	0000	0000
48	F0F1	0000	0113	0113	0000	0000	FF00	007F
50	6A50	6576	A5A6	B0A2	0433	2426	2737	2706
58	3802	000E	0000	0000	3333	0333	0333	0333
60	001F	0000	0000	0000	0000	0000	0000	0000
68	3321	3333	0213	1202	0023	0000	0000	0000
70	1221	1222	3233	3131	0032	0000	0000	1F1F
78	1F1F	0000	0000	0200	8600	0007	C0C0	C0C0